DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-300-03 — Internal inspection of horizontal drum D-101

| | | | |
|---|---|---|---|
| **Area** | A300 Exchangers and drum | **Task** | VI Internal inspection |
| **Priority** | 2 (integrity) | **Issued by** | Facility owner (demonstration) |
| **Primary asset** | D-101 — D-101 horizontal vessel 60" ID × 180" — E 84'-0"  N -25'-0"  EL 107'-6" | | |
| **Equipment** | D-101 | **Lines** | 6"-P300-P-1012, 4"-P300-P-1013, 1"-P150-V-6016, 1"-P150-D-6017, 1"-P150-N-3014, 3/4"-P300-I-6018, 3/4"-P300-I-6019, 3/4"-P300-SP-6020 |
| **Isometrics** | ISO-P-1012, ISO-P-1013, ISO-V-6016, ISO-D-6017, ISO-N-3014, ISO-I-6018, ISO-I-6019, ISO-SP-6020 (owner's drawings, attached) | | |
| **Permits** | General work permit, Confined-space entry permit | | |

## Reason

Scheduled internal inspection of the drum.

## Scope of work

1. Isolate D-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 14 TMLs.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: D-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

## Owner's isometric drawings (SW-IMP-6)

| Drawing | Line drawn | Work line | Sheets | Match to the plant model |
|---|---|---|---|---|
| ISO-P-1012 | 6"-P300-P-1012 | 6"-P300-P-1012 | 2 | agrees with the model |
| ISO-P-1013 | 4"-P300-P-1013 | 4"-P300-P-1013 | 5 | agrees with the model |
| ISO-V-6016 | 1"-P150-V-6016 | 1"-P150-V-6016 | 1 | agrees with the model |
| ISO-D-6017 | 1"-P150-D-6017 | 1"-P150-D-6017 | 1 | agrees with the model |
| ISO-N-3014 | 1"-P150-N-3014 | 1"-P150-N-3014 | 1 | agrees with the model |
| ISO-I-6018 | 3/4"-P300-I-6018 | 3/4"-P300-I-6018 | 1 | agrees with the model |
| ISO-I-6019 | 3/4"-P300-I-6019 | 3/4"-P300-I-6019 | 1 | agrees with the model |
| ISO-SP-6020 | 3/4"-P300-SP-6020 | 3/4"-P300-SP-6020 | 1 | agrees with the model |

ISO-P-1012 (PDF): ../isometrics-imported/ISO-P-1012.pdf · ISO-P-1013 (PDF): ../isometrics-imported/ISO-P-1013.pdf · ISO-V-6016 (PDF): ../isometrics-imported/ISO-V-6016.pdf · ISO-D-6017 (PDF): ../isometrics-imported/ISO-D-6017.pdf · ISO-N-3014 (PDF): ../isometrics-imported/ISO-N-3014.pdf · ISO-I-6018 (PDF): ../isometrics-imported/ISO-I-6018.pdf · ISO-I-6019 (PDF): ../isometrics-imported/ISO-I-6019.pdf · ISO-SP-6020 (PDF): ../isometrics-imported/ISO-SP-6020.pdf — the sheets that hold the work follow the ITP.

## Assets in the plant model (SW-WO-7)

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| D-101 | equipment | D-101 horizontal vessel 60" ID × 180" | | E 84'-0"  N -25'-0"  EL 107'-6" |
| 6"-P300-P-1012/B2 | item | VALVE, GATE, WEDGE, OS&Y 6" P300 | 6"-P300-P-1012 | E 86'-6"  N -25'-0"  EL 111'-3-7/16" |
| D-101-N1 | nozzle | Nozzle D-101-N1 (inlet, 6" P300) on D-101 | | E 86'-6"  N -25'-0"  EL 110'-8-5/16" |
| 4"-P300-P-1013/A2 | item | VALVE, GATE, WEDGE, OS&Y 4" P300 | 4"-P300-P-1013 | E 98'-0"  N -25'-0"  EL 103'-8-3/16" |
| D-101-N2 | nozzle | Nozzle D-101-N2 (outlet, 4" P300) on D-101 | | E 98'-0"  N -25'-0"  EL 104'-3-15/16" |
| 3"-P300-RV-4001/2 | item | VALVE, PRESSURE SAFETY VALVE 3" P300 INLET × 4" P150 OUTLET | 3"-P300-RV-4001 | E 94'-0"  N -25'-0"  EL 111'-6" |
| 1"-P150-V-6016/2 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-V-6016 | E 89'-0"  N -25'-0"  EL 111'-1/2" |
| D-101-N4 | nozzle | Nozzle D-101-N4 (vent, 1" P150) on D-101 | | E 89'-0"  N -25'-0"  EL 110'-5-11/16" |
| 1"-P150-D-6017/2 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-D-6017 | E 85'-6"  N -25'-0"  EL 103'-11-1/2" |
| D-101-N5 | nozzle | Nozzle D-101-N5 (drain, 1" P150) on D-101 | | E 85'-6"  N -25'-0"  EL 104'-6-3/8" |
| 1"-P150-N-3014/B2 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-N-3014 | E 91'-6"  N -25'-0"  EL 111'-1/2" |
| D-101-N6 | nozzle | Nozzle D-101-N6 (nitrogen, 1" P150) on D-101 | | E 91'-6"  N -25'-0"  EL 110'-5-11/16" |
| 3/4"-P300-I-6018/2 | item | VALVE, BALL, FLOATING 3/4" P300 | 3/4"-P300-I-6018 | E 91'-6"  N -28'-6-1/4"  EL 107'-6" |
| D-101-N7 | nozzle | Nozzle D-101-N7 (instrument, 0.75" P300) on D-101 | | E 91'-6"  N -27'-11-1/2"  EL 107'-6" |
| 3/4"-P300-I-6019/2 | item | VALVE, BALL, FLOATING 3/4" P300 | 3/4"-P300-I-6019 | E 95'-0"  N -28'-6-1/4"  EL 107'-6" |

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| D-101-N8 | nozzle | Nozzle D-101-N8 (instrument, 0.75" P300) on D-101 | | E 95'-0"  N -27'-11-1/2"  EL 107'-6" |
| 3/4"-P300-SP-6020/2 | item | VALVE, BALL, FLOATING 3/4" P300 | 3/4"-P300-SP-6020 | E 93'-0"  N -25'-0"  EL 103'-11-3/4" |
| D-101-N9 | nozzle | Nozzle D-101-N9 (sample, 0.75" P300) on D-101 | | E 93'-0"  N -25'-0"  EL 104'-6-1/2" |

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | valve | 6"-P300-P-1012/B2 | 6"-P300-P-1012 | Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1012/B2) on D-101-N1 | E 86'-6"  N -25'-0"  EL 111'-3-7/16" | | |
| 2 | blind | D-101-N1 | 6"-P300-P-1012 | Fit a blind between 6"-P300-P-1012/B2 and the nozzle flange D-101-N1 | E 86'-6"  N -25'-0"  EL 110'-8-5/16" | | |
| 3 | valve | 4"-P300-P-1013/A2 | 4"-P300-P-1013 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1013/A2) on D-101-N2 | E 98'-0"  N -25'-0"  EL 103'-8-3/16" | | |
| 4 | blind | D-101-N2 | 4"-P300-P-1013 | Fit a blind between 4"-P300-P-1013/A2 and the nozzle flange D-101-N2 | E 98'-0"  N -25'-0"  EL 104'-3-15/16" | | |
| 5 | removal | 3"-P300-RV-4001/2 | 3"-P300-RV-4001 | Remove the relief valve on D-101-N3 and blind the nozzle | | | |
| 6 | valve | 1"-P150-V-6016/2 | 1"-P150-V-6016 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6016/2) on D-101-N4 | E 89'-0"  N -25'-0"  EL 111'-1/2" | | |
| 7 | blind | D-101-N4 | 1"-P150-V-6016 | Fit a blind between 1"-P150-V-6016/2 and the nozzle flange D-101-N4 | E 89'-0"  N -25'-0"  EL 110'-5-11/16" | | |
| 8 | valve | 1"-P150-D-6017/2 | 1"-P150-D-6017 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6017/2) on D-101-N5 | E 85'-6"  N -25'-0"  EL 103'-11-1/2" | | |
| 9 | blind | D-101-N5 | 1"-P150-D-6017 | Fit a blind between 1"-P150-D-6017/2 and the nozzle flange D-101-N5 | E 85'-6"  N -25'-0"  EL 104'-6-3/8" | | |
| 10 | valve | 1"-P150-N-3014/B2 | 1"-P150-N-3014 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3014/B2) on D-101-N6 | E 91'-6"  N -25'-0"  EL 111'-1/2" | | |
| 11 | blind | D-101-N6 | 1"-P150-N-3014 | Fit a blind between 1"-P150-N-3014/B2 and the nozzle flange D-101-N6 | E 91'-6"  N -25'-0"  EL 110'-5-11/16" | | |
| 12 | valve | 3/4"-P300-I-6018/2 | 3/4"-P300-I-6018 | Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6018/2) on D-101-N7 | E 91'-6"  N -28'-6-1/4"  EL 107'-6" | | |
| 13 | blind | D-101-N7 | 3/4"-P300-I-6018 | Fit a blind between 3/4"-P300-I-6018/2 and the nozzle flange D-101-N7 | E 91'-6"  N -27'-11-1/2"  EL 107'-6" | | |
| 14 | valve | 3/4"-P300-I-6019/2 | 3/4"-P300-I-6019 | Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6019/2) on D-101-N8 | E 95'-0"  N -28'-6-1/4"  EL 107'-6" | | |
| 15 | blind | D-101-N8 | 3/4"-P300-I-6019 | Fit a blind between 3/4"-P300-I-6019/2 and the nozzle flange D-101-N8 | E 95'-0"  N -27'-11-1/2"  EL 107'-6" | | |
| 16 | valve | 3/4"-P300-SP-6020/2 | 3/4"-P300-SP-6020 | Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-SP-6020/2) on D-101-N9 | E 93'-0"  N -25'-0"  EL 103'-11-3/4" | | |
| 17 | blind | D-101-N9 | 3/4"-P300-SP-6020 | Fit a blind between 3/4"-P300-SP-6020/2 and the nozzle flange D-101-N9 | E 93'-0"  N -25'-0"  EL 104'-6-1/2" | | |

## Inspections

| Id | Inspection | Criteria | Basis |
|---|---|---|---|
| WO-300-03-I1 | Internal visual examination of the shell, heads and nozzles | Findings recorded against the owner's integrity criteria | SW-NDE-4 |
| WO-300-03-I2 | Ultrasonic thickness at the 14 TMLs listed | Each reading above the owner's minimum wall | SW-WO-6 |

## Thickness monitoring locations (SW-WO-6)

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-300-03-TML01 | D-101 | Nozzle neck D-101-N1 | E 86'-6"  N -25'-0"  EL 110'-8-5/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-03-TML02 | D-101 | Nozzle neck D-101-N2 | E 98'-0"  N -25'-0"  EL 104'-3-15/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-03-TML03 | D-101 | Nozzle neck D-101-N3 | E 94'-0"  N -25'-0"  EL 110'-10-5/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-03-TML04 | D-101 | Nozzle neck D-101-N4 | E 89'-0"  N -25'-0"  EL 110'-5-11/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-03-TML05 | D-101 | Nozzle neck D-101-N5 | E 85'-6"  N -25'-0"  EL 104'-6-3/8" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-03-TML06 | D-101 | Nozzle neck D-101-N6 | E 91'-6"  N -25'-0"  EL 110'-5-11/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-300-03-TML07 | D-101 | Nozzle neck D-101-N7 | E 91'-6"  N -27'-11-1/2"  EL 107'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-03-TML08 | D-101 | Nozzle neck D-101-N8 | E 95'-0"  N -27'-11-1/2"  EL 107'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-03-TML09 | D-101 | Nozzle neck D-101-N9 | E 93'-0"  N -25'-0"  EL 104'-6-1/2" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-03-TML10 | D-101 | Left head centre | E 82'-8-3/4"  N -25'-0"  EL 107'-6" | | centre: ____ | | | |
| WO-300-03-TML11 | D-101 | Right head centre | E 100'-3-1/4"  N -25'-0"  EL 107'-6" | | centre: ____ | | | |
| WO-300-03-TML12 | D-101 | Shell at 0.000" from the left tangent | E 84'-0"  N -25'-0"  EL 107'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-300-03-TML13 | D-101 | Shell at 6'-0" from the left tangent | E 90'-0"  N -25'-0"  EL 107'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-300-03-TML14 | D-101 | Shell at 12'-0" from the left tangent | E 96'-0"  N -25'-0"  EL 107'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |

## Test — Service leak test

| | |
|---|---|
| **Description** | Every opened joint checked at operating pressure on start-up |
| **Basis** | SW-TEST-2 |

## Inspection and test plan (SW-ITP-1..3)

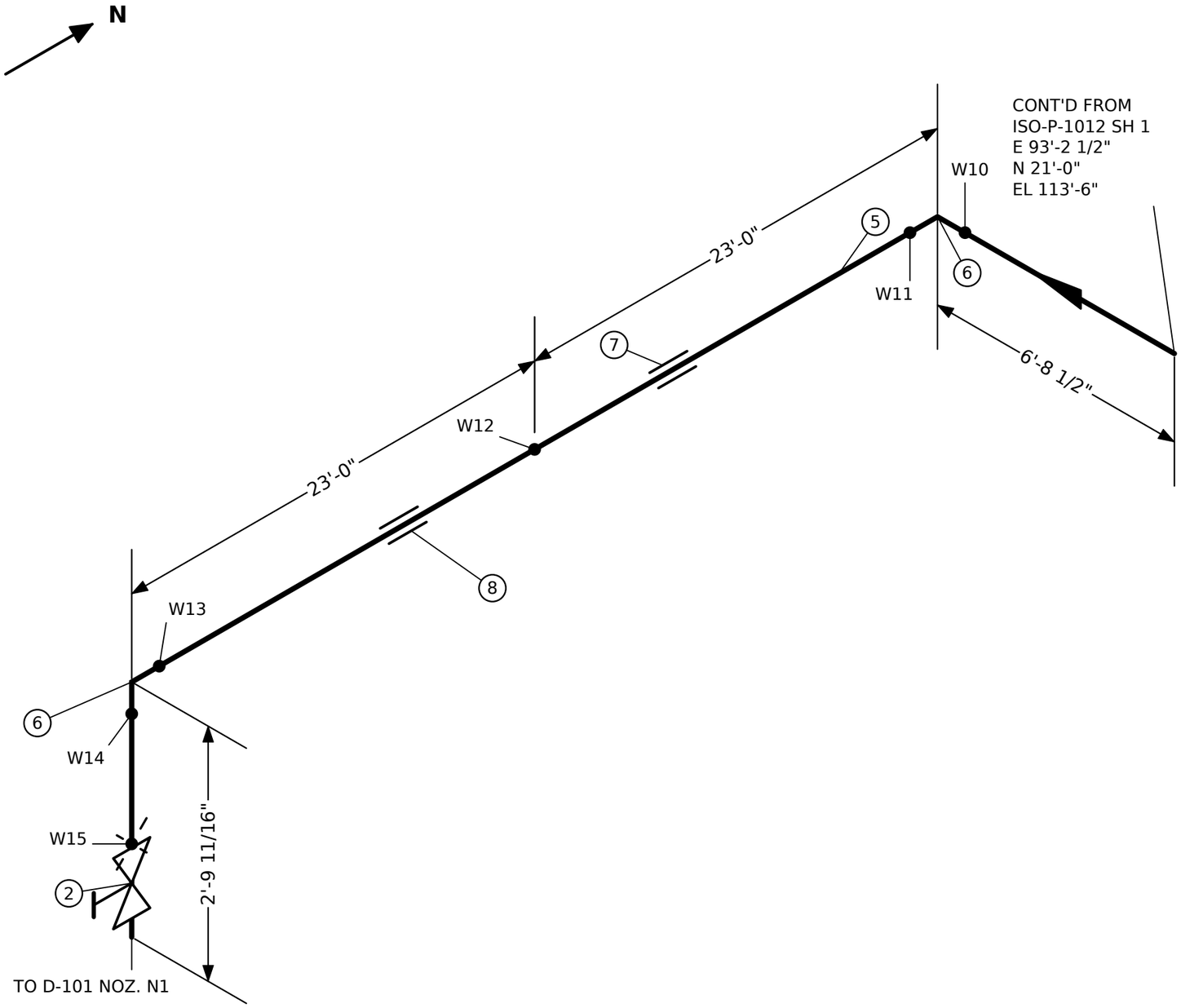
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Equipment vented, purged and gas-tested for entry; confined-space entry permit issued | Owner entry procedure | Atmosphere safe for entry per owner limits | Gas test record, entry permit | H | H | — | | | | |
| 3 | Opened for entry; cleaned for inspection | Owner procedure | Surfaces clean enough to inspect | Daily report | S | S | — | | | | |
| 4 | Internal visual examination | SW-NDE-4 | Findings recorded | Inspection report with photographs | W | H | W | | | | |
| 5 | Ultrasonic thickness at 14 TMLs | SW-WO-6 | Above the owner's minimum wall | Thickness record per TML | S | R | R | | | | |
| 6 | Owner's decision on findings (repair, monitor or accept) | Owner integrity procedure | Decision recorded | Signed disposition | R | H | R | | | | |
| 7 | Closed with new gaskets on every opened joint; studs to preload | SW-WO-5 | Preload as stated | Bolt-up record | W | W | — | | | | |
| 8 | Service leak test of opened joints on start-up | SW-TEST-2 | No leak | Leak test record | W | W | — | | | | |
| 9 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 10 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

## Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

| Tag | Role | Drawing | Sheet | Weld label |
|---|---|---|---|---|
| 6"-P300-P-1012/B2 | isolation: valve | ISO-P-1012 | 2 | |
| 4"-P300-P-1013/A2 | isolation: valve | ISO-P-1013 | 1 | |
| 3"-P300-RV-4001/2 | isolation: removal | ISO-RV-4001 | 1 | |
| 1"-P150-V-6016/2 | isolation: valve | ISO-V-6016 | 1 | |
| 1"-P150-D-6017/2 | isolation: valve | ISO-D-6017 | 1 | |
| 1"-P150-N-3014/B2 | isolation: valve | ISO-N-3014 | 1 | |
| 3/4"-P300-I-6018/2 | isolation: valve | ISO-I-6018 | 1 | |
| 3/4"-P300-I-6019/2 | isolation: valve | ISO-I-6019 | 1 | |
| 3/4"-P300-SP-6020/2 | isolation: valve | ISO-SP-6020 | 1 | |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 2 | 6" | GASKET 6" P300 COMPRESSED SHEET |
| 2 | 1 | 6" | VALVE, GATE, WEDGE, OS&Y 6" P300 |
| 3 | 2 | 6" | STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300 |
| 4 | 1 | 6" | FLANGE WELD-NECK 6" P300 RAISED FACE |
| 5 | 50'-8" | 6" | PIPE 6" P300 BEVELLED ENDS |
| 6 | 2 | 6" | ELBOW 90° R = 5/3 OD 6" P300 |
| 7 | 1 | 6" | HANGER FOR 6" LINE (HANGER-6x231.0) |
| 8 | 1 | 6" | HANGER FOR 6" LINE (HANGER-6x23.0) |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

COMPANY

**SMARTWELD**

TITLE

**HYDROCARBON PROCESS**

**NOTES:**

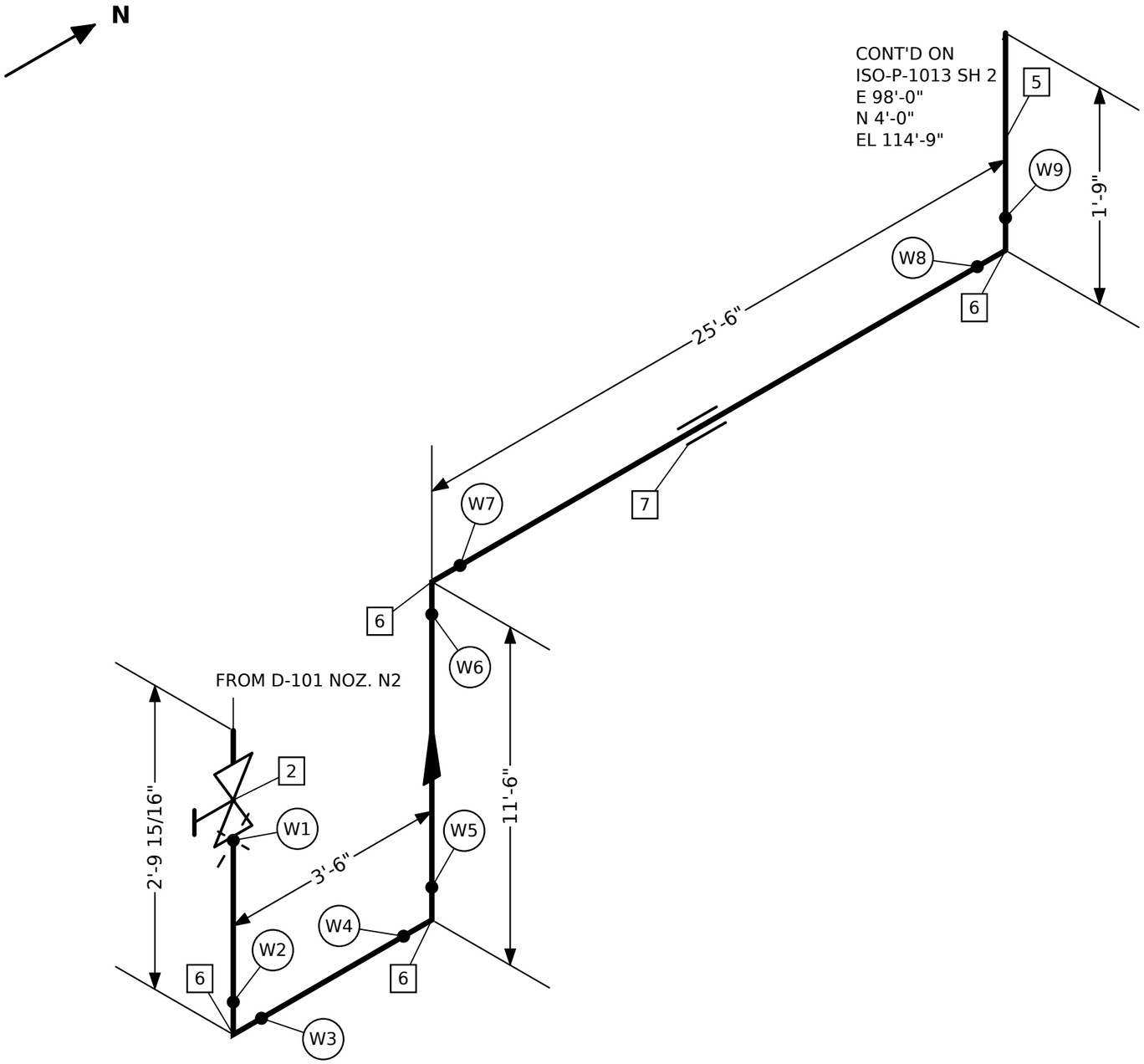
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | NONE | - |
| PWHT | NDE | FLUID |
| - | - | P |

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 6"-P300-P-1012 | H300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-P-1012 | 2 OF 2 | 1 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 2 | 4" | GASKET 4" P300 COMPRESSED SHEET |
| 2 | 1 | 4" | VALVE, GATE, WEDGE, OS&Y 4" P300 |
| 3 | 2 | 4" | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300 |
| 4 | 1 | 4" | FLANGE WELD-NECK 4" P300 RAISED FACE |
| 5 | 38'-10 11/16" | 4" | PIPE 4" P300 BEVELLED ENDS |
| 6 | 4 | 4" | ELBOW 90° R = 5/3 OD 4" P300 |
| 7 | 1 | 4" | HANGER FOR 4" LINE (HANGER-4x29.0) |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | NONE | - |
| PWHT | NDE | FLUID |
| - | - | P |

COMPANY

**SMARTWELD**

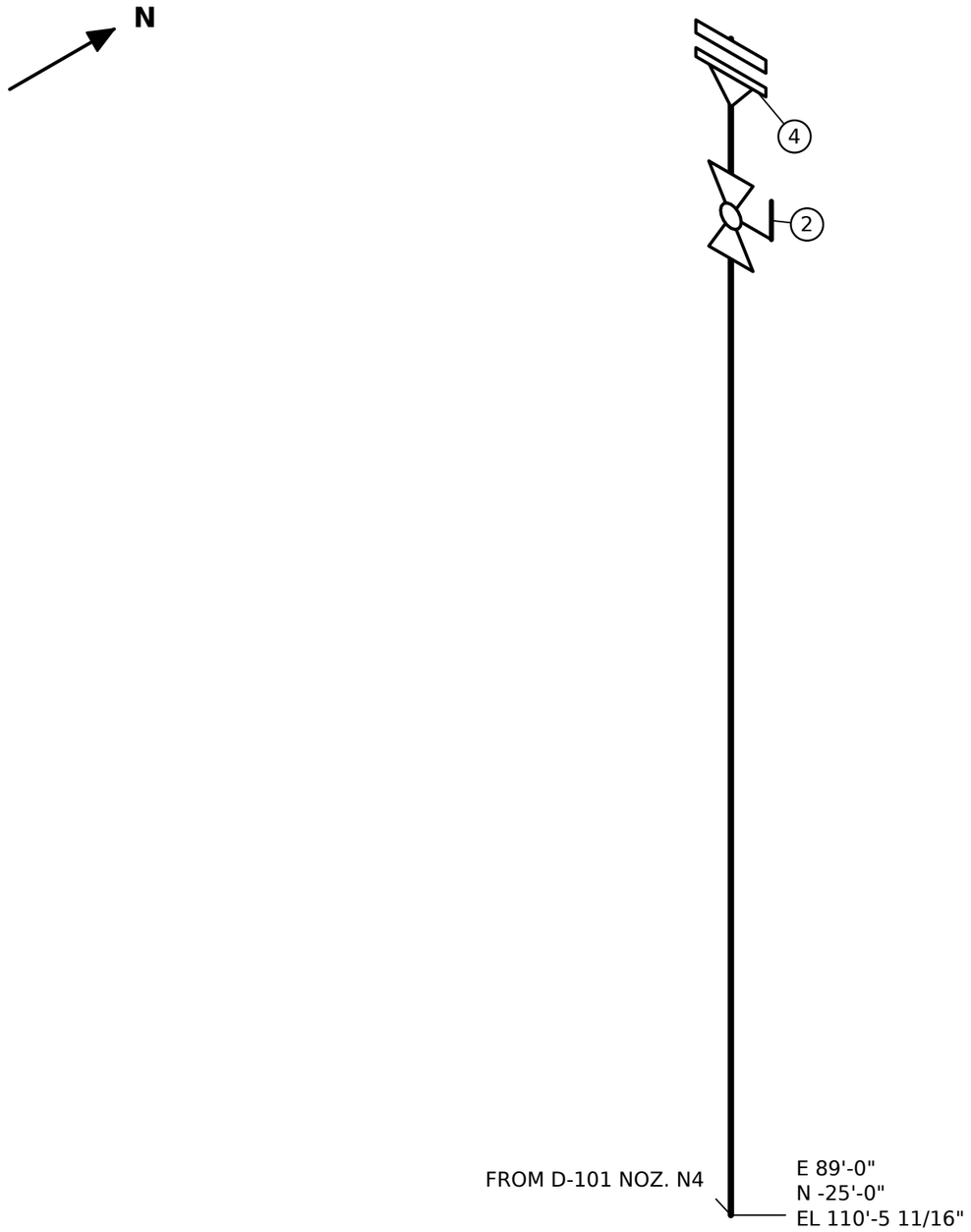
TITLE

**HYDROCARBON PROCESS**

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 4"-P300-P-1013 | H300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-P-1013 | 1 OF 5 | 1 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 2 | 1" | GASKET 1" P150 COMPRESSED SHEET |
| 2 | 1 | 1" | VALVE, BALL, FLOATING 1" P150 |
| 3 | 2 | 1" | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 |
| 4 | 1 | 1" | FLANGE BLIND 1" P150 RAISED FACE |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS
4. HYDROTEST BEFORE PAINTING

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | NONE | - |

| PWHT | NDE | FLUID |
|---|---|---|
| - | - | V |

COMPANY

**SMARTWELD**

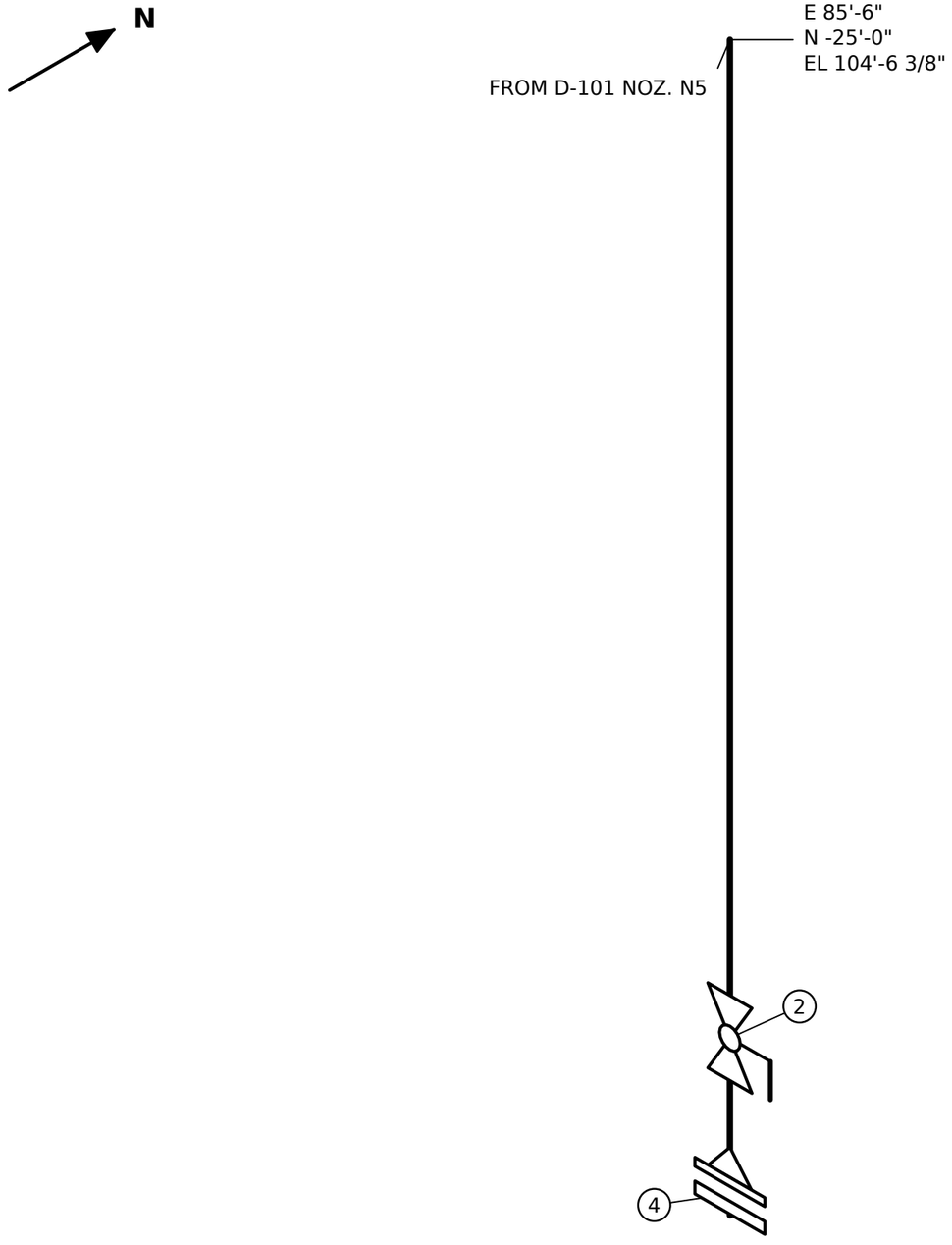
TITLE

**HYDROCARBON PROCESS**

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 1"-P150-V-6016 | H150 | 150 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-V-6016 | 1 OF 1 | 1 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 2 | 1" | GASKET 1" P150 COMPRESSED SHEET |
| 2 | 1 | 1" | VALVE, BALL, FLOATING 1" P150 |
| 3 | 2 | 1" | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 |
| 4 | 1 | 1" | FLANGE BLIND 1" P150 RAISED FACE |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

| | | | |
|---|---|---|---|
| TEST PRESS.<br>- | INSULATION<br>**NONE** | PAINT<br>- | |
| PWHT<br>- | NDE<br>- | FLUID<br>**D** | |

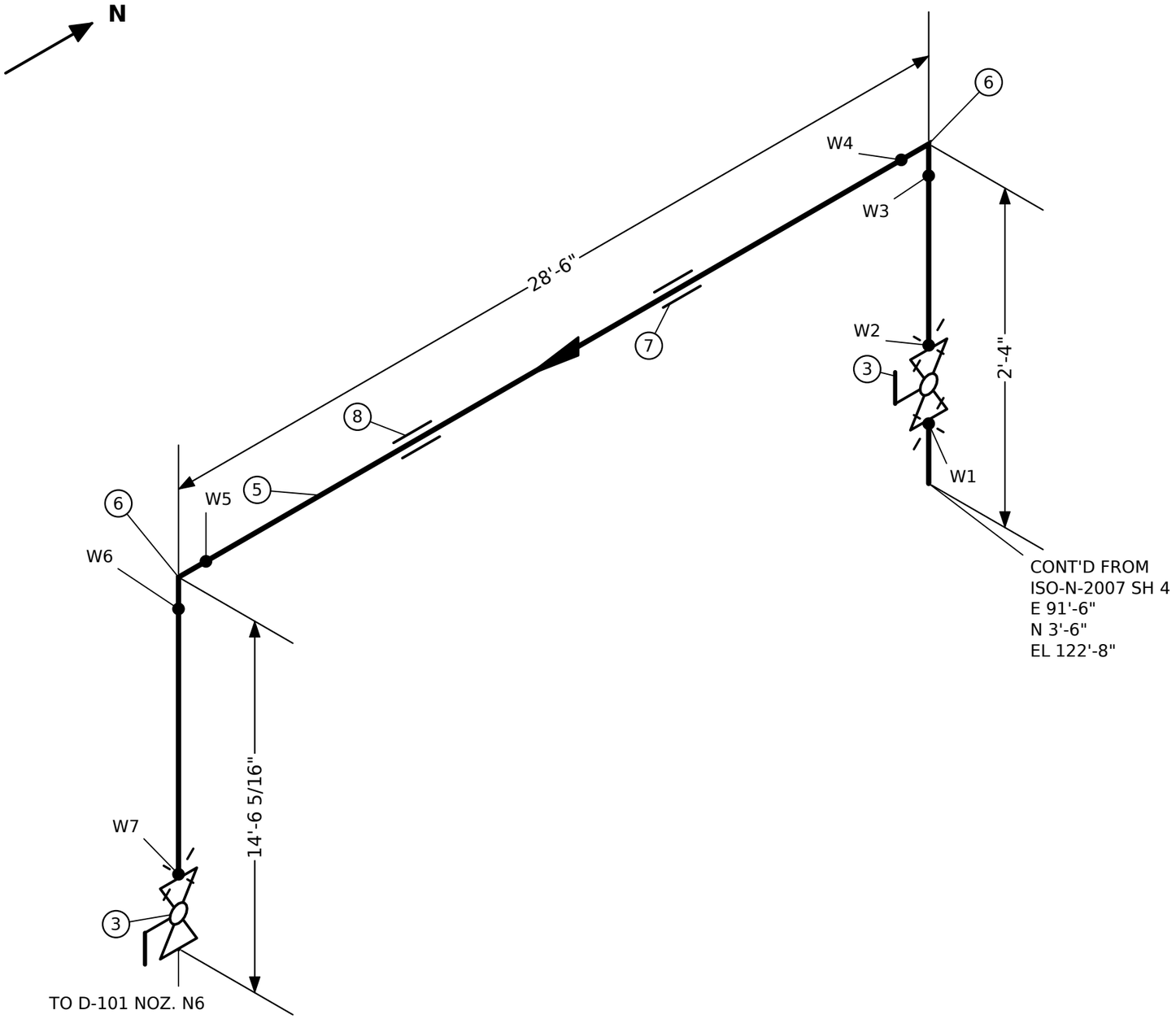
COMPANY

**SMARTWELD**

TITLE

**HYDROCARBON PROCESS**

| | | | |
|---|---|---|---|
| LINE NO.<br>**1"-P150-D-6017** | PIPING CLASS<br>**H150** | DESIGN PRESS.<br>**150 PSIG** | DESIGN TEMP.<br>**-** |
| DRAWING NO.<br>**ISO-D-6017** | | SHEET<br>**1 OF 1** | REV<br>**1** |
| DATE<br>**2026-11-01** | DRAWN<br>**-** | CHECKED<br>**-** | SCALE<br>**NTS** |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 3 | 1" | FLANGE WELD-NECK 1" P150 RAISED FACE |
| 2 | 4 | 1" | GASKET 1" P150 COMPRESSED SHEET |
| 3 | 2 | 1" | VALVE, BALL, FLOATING 1" P150 |
| 4 | 4 | 1" | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 |
| 5 | 41'-5 7/8" | 1" | PIPE 1" P150 BEVELLED ENDS |
| 6 | 2 | 1" | ELBOW 90° R = 5/3 OD 1" P150 |
| 7 | 1 | 1" | HANGER FOR 1" LINE (HANGER-1x93.0) |
| 8 | 1 | 1" | HANGER FOR 1" LINE (HANGER-1x83.0) |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

**NOTES:**
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | NONE | - |
| **PWHT** | **NDE** | **FLUID** |
| - | - | N |

COMPANY

**SMARTWELD**

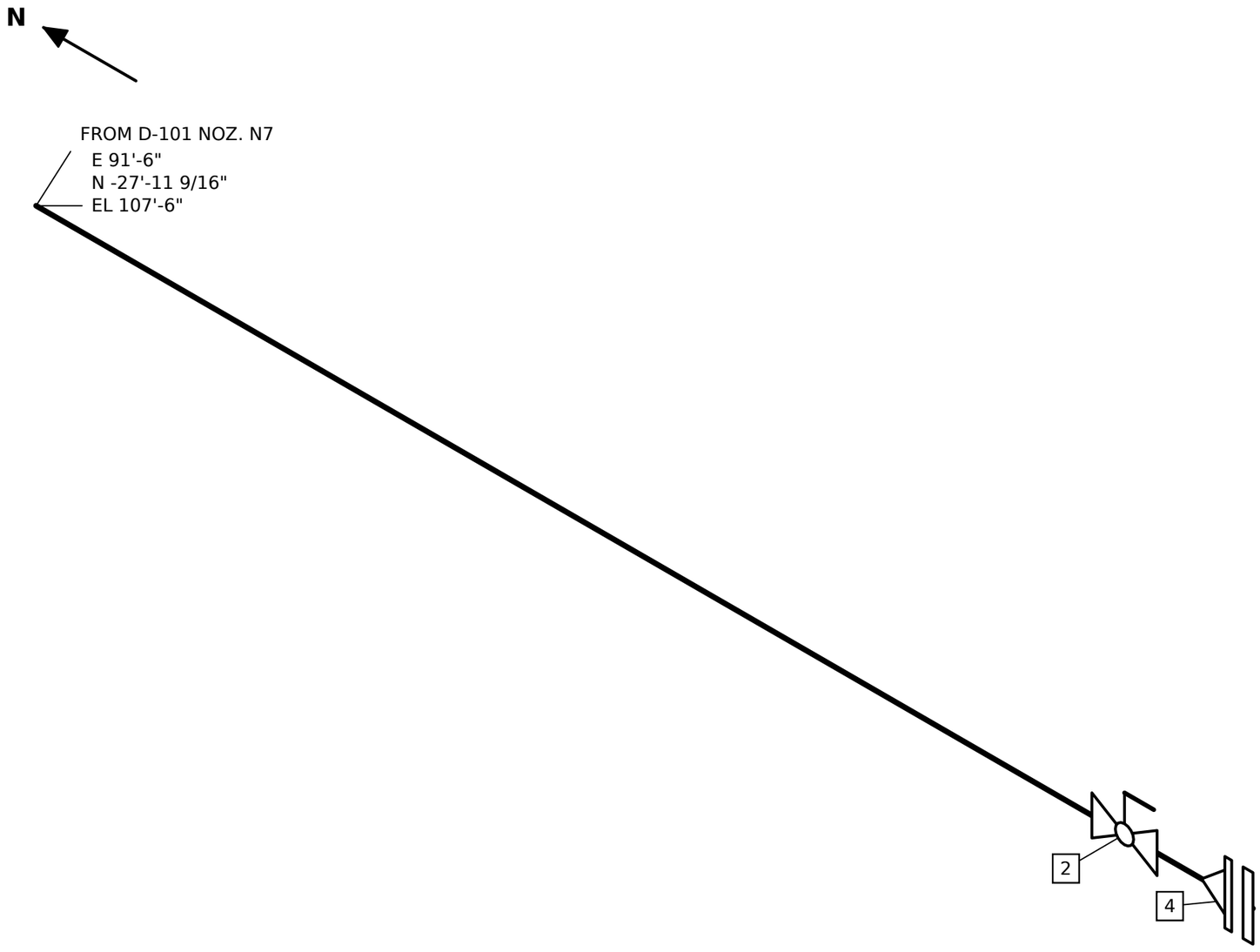
TITLE

**AIR AND GASES**

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 1"-P150-N-3014 | G150 | 150 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-N-3014 | 1 OF 1 | 1 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 2 | 3/4" | GASKET 3/4" P300 COMPRESSED SHEET |
| 2 | 1 | 3/4" | VALVE, BALL, FLOATING 3/4" P300 |
| 3 | 2 | 3/4" | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 |
| 4 | 1 | 3/4" | FLANGE BLIND 3/4" P300 RAISED FACE |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS

| | | |
|---|---|---|
| TEST PRESS.<br>- | INSULATION<br>NONE | PAINT<br>- |
| PWHT<br>- | NDE<br>- | FLUID<br>I |

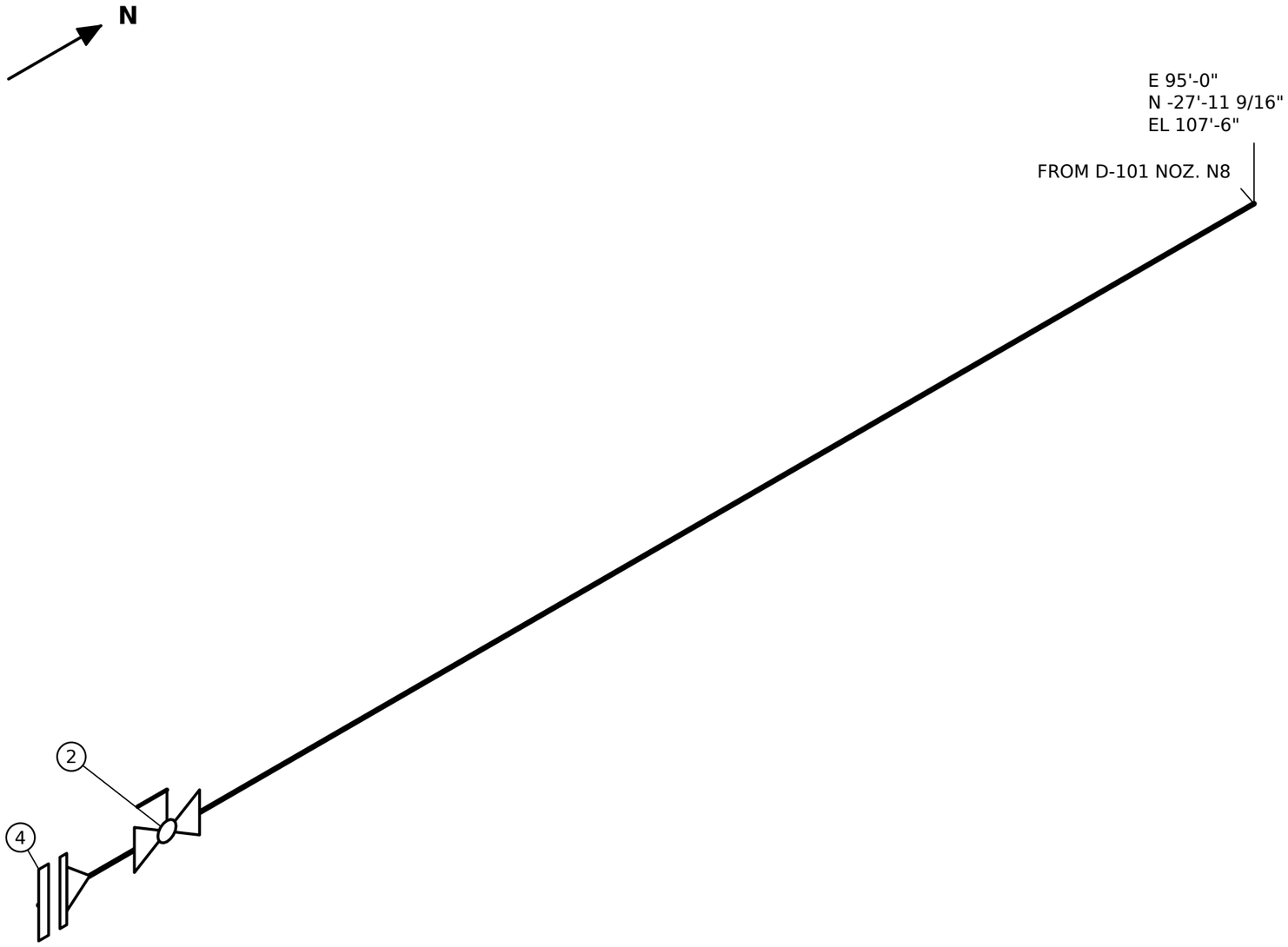
COMPANY

**SMARTWELD**

TITLE

**HYDROCARBON PROCESS**

| | | | |
|---|---|---|---|
| LINE NO.<br>3/4"-P300-I-6018 | PIPING CLASS<br>H300 | DESIGN PRESS.<br>300 PSIG | DESIGN TEMP.<br>- |
| DRAWING NO.<br>ISO-I-6018 | | SHEET<br>1 OF 1 | REV<br>1 |
| DATE<br>2026-11-01 | DRAWN<br>- | CHECKED<br>- | SCALE<br>NTS |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 2 | 3/4" | GASKET 3/4" P300 COMPRESSED SHEET |
| 2 | 1 | 3/4" | VALVE, BALL, FLOATING 3/4" P300 |
| 3 | 2 | 3/4" | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 |
| 4 | 1 | 3/4" | FLANGE BLIND 3/4" P300 RAISED FACE |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | NONE | - |
| PWHT | NDE | FLUID |
| - | - | I |

COMPANY

**SMARTWELD**

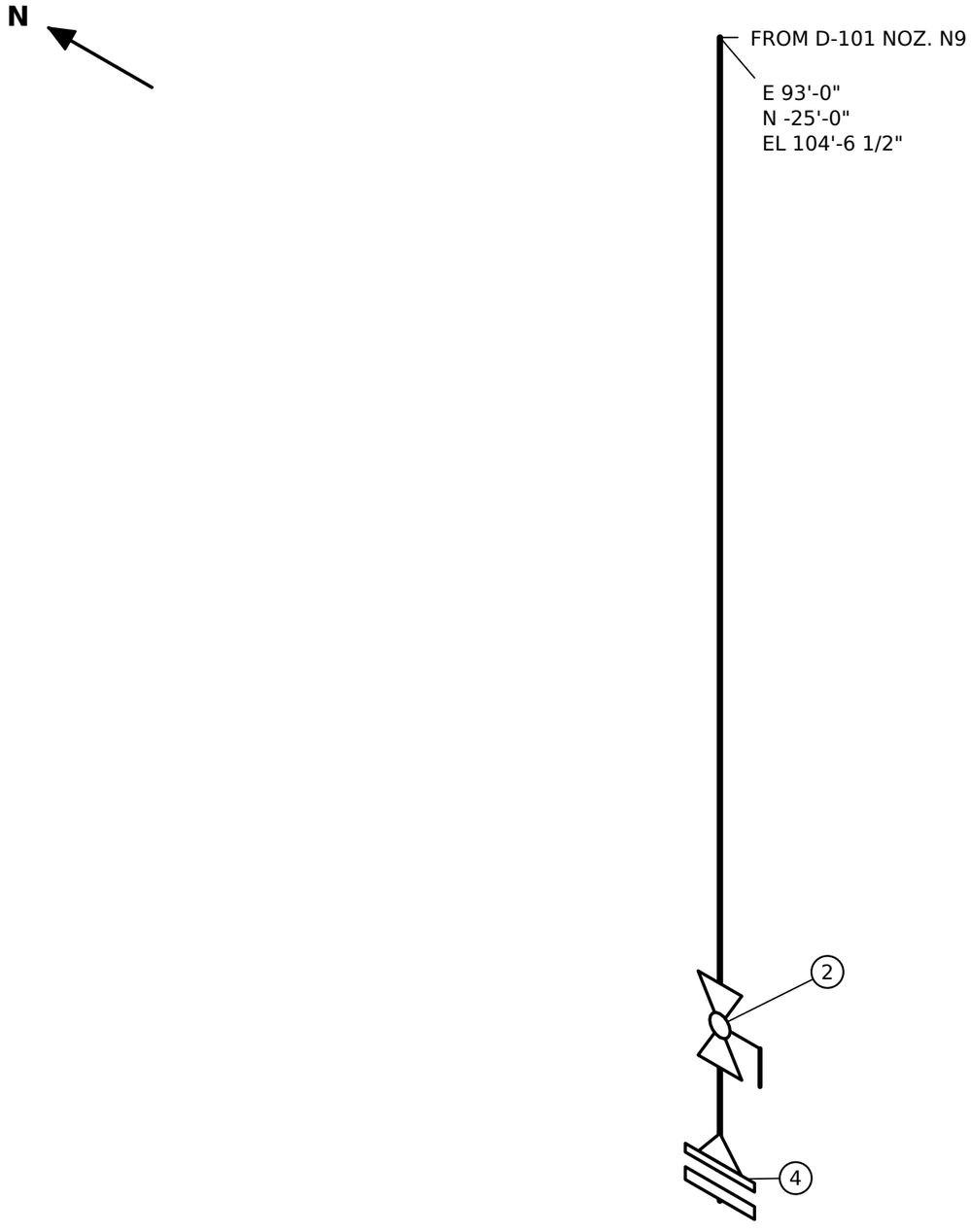
TITLE

**HYDROCARBON PROCESS**

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 3/4"-P300-I-6019 | H300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-I-6019 | 1 OF 1 | 1 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 2 | 3/4" | GASKET 3/4" P300 COMPRESSED SHEET |
| 2 | 1 | 3/4" | VALVE, BALL, FLOATING 3/4" P300 |
| 3 | 2 | 3/4" | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 |
| 4 | 1 | 3/4" | FLANGE BLIND 3/4" P300 RAISED FACE |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

COMPANY

**SMARTWELD**

TITLE

**HYDROCARBON PROCESS**

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. REMOVE BURRS FROM ALL CUT ENDS

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | NONE | - |

| PWHT | NDE | FLUID |
|---|---|---|
| - | - | SP |

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 3/4"-P300-SP-6020 | H300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-SP-6020 | 1 OF 1 | 1 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |

# Notices

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